THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · NCERT-TYPE CHAPTER TEST SET 2

Chapter 6: **Perimeter and Area** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The area of a circle with circumference 132 cm (take $\pi = \frac{22}{7}$) is (cm^2): 1

(a) 1388 **(b)** 1385 **(c)** 1387 **(d)** 1386

2. A wheel of radius 35 cm (take $\pi = \frac{22}{7}$) makes one full turn. The distance covered (in cm) is: 1

(a) 221 **(b)** 219 **(c)** 222 **(d)** 220

3. The area of a rhombus with diagonals 16 cm and 12 cm is (cm^2): 1

(a) 98 **(b)** 96 **(c)** 97 **(d)** 95

4. The area of a cyclic quadrilateral with sides 4 cm, 8 cm, 14 cm and 22 cm is (cm^2): 1

(a) 81 **(b)** 82 **(c)** 80 **(d)** 79

5. The length of an arc of angle 90° in a circle of radius 14 cm (take $\pi = \frac{22}{7}$) is (cm): 1

(a) 21 **(b)** 23 **(c)** 24 **(d)** 22

6. A square has the same perimeter as a circle of circumference 176 cm. The side of the square (cm) is: 1

(a) 44 **(b)** 43 **(c)** 45 **(d)** 46

7. The area of a triangle with sides 9 cm, 10 cm and 17 cm is (cm^2): 1

(a) 38 **(b)** 35 **(c)** 37 **(d)** 36

8. The semi-perimeter of a triangle with sides 10 cm, 10 cm and 12 cm is (cm): 1

(a) 15 **(b)** 18 **(c)** 17 **(d)** 16

9. The area of a circle with circumference 220 cm (take $\pi = \frac{22}{7}$) is (cm^2): 1

(a) 3850 **(b)** 3852 **(c)** 3851 **(d)** 3849

10. A wheel of radius 42 cm (take $\pi = \frac{22}{7}$) makes one full turn. The distance covered (in cm) is: 1

(a) 264 **(b)** 263 **(c)** 265 **(d)** 266

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A wheel of radius 35 cm covers a distance of 8800 cm. How many revolutions does it make (take $\pi = \frac{22}{7}$)? 2

12. Find the length of an arc of 90° in a circle of radius 14 cm (take $\pi = \frac{22}{7}$). 2

13. Find the area of a triangle with sides 9 cm, 10 cm, 17 cm. 2

14. A wheel of radius 21 cm covers a distance of 6600 cm. How many revolutions does it make (take $\pi = \frac{22}{7}$)? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the area of the triangle with sides 9 cm, 10 cm and 17 cm, and the length of the altitude on the side of 17 cm. 3

16. Find the area of the quadrilateral with sides 8, 10, 14, 20 cm that is inscribed in a circle. 3

17. Find the area of a parallelogram with base 22 cm and height 5 cm, and the base of a triangle of height 10 cm having the same area. 3

OR

Find the area of the triangle with sides 7 cm, 15 cm and 20 cm, and the length of the altitude on the side of 20 cm. 3

18. Find the area of the quadrilateral with sides 5, 14, 19, 20 cm that is inscribed in a circle. 3

OR

Find the area of a parallelogram with base 16 cm and height 12 cm, and the base of a triangle of height 24 cm having the same area. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A circular track has inner radius 28 m and outer radius 42 m (take $\pi = \frac{22}{7}$). Find the area of the track and the cost of levelling it at ₹3 per m^2. 5

OR

A rectangular lawn 50 m $\times$ 30 m has a path 2 m wide all around it, outside the lawn. Find the area of the path. 5

20. A circular track has inner radius 35 m and outer radius 42 m (take $\pi = \frac{22}{7}$). Find the area of the track and the cost of levelling it at ₹6 per m^2. 5

OR

A rectangular lawn 36 m $\times$ 24 m has a path 4 m wide all around it, outside the lawn. Find the area of the path. 5

END OF QUESTION PAPER — Total: 40 Marks

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